Jian Feng

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EDUCATION

University of Waterloo 09/2023 – 04/2028

Bachelor of Computing and Financial Management, Honours, Co-op

Majors: Computer Science, Finance, Statistics

EXPERIENCE

Manulife 05/2026 – 08/2026

Co-op

- Triage ~10 weekly cloud/network infra tickets from global ETS teams via ServiceNow, owning resolution and stakeholder follow-through.
- Built `aks-onboarding-automation` replacing manual ServiceNow ticket exchanges; projected to cut onboarding from days to minutes.

Treasury Board of Canada Secretariat 09/2025 – 12/2025

Co-op

- Shipped a .NET MVC module that auto-normalized job titles across 5K public firms via pattern matching; cut manual data cleaning 80%.
- Migrated 30 test cases across 5 suites from Selenium to Katalon and stood up CI/CD on Azure DevOps, presenting demo to executives.
- Refactored core modules of an enterprise data platform and tuned SQL queries, cutting page load 30% and improving maintainability.

Lansi Institute of Artificial Intelligence in Medicine 04/2024 – 12/2024

Co-op (Remote)

- Built genomic ETL pipelines for ~100GB multi-omics data; per-sample fan-out via Python `multiprocessing` cut runtime 89%.
- Co-authored published T2D drug-target meta-analysis (68 studies); trained XGBoost, SMR, MR^2 to link variants to drug response.

Sanyang Technology Co., Ltd. 07/2024 – 11/2024

Intern

- Built Gantt visualizer of a flexible job-shop scheduling system, unblocking algorithm A/B runs on benchmark instances.
- Shipped Spring Boot CRUD APIs for cloud reporting/BI modules, unblocking front-end dashboards and downstream ETL consumers.

PROJECTS

wlp4-compiler

- Multi-pass C++ pipeline: `wlp4scan` → `wlp4parse` (LR) → `wlp4type` → `wlp4gen` emitting ARM64 assembly.
- Codegen lowers control flow, procedures, pointer/array ops with a stable calling convention; transitions traceable via `.wlp4ti` trees.
- −79.6% output size vs the reference `wlp4c` across 65 programs; CI gates every push on the 25-program byte-equal regression suite.

MiniKV

- Go LSM engine — CRC32C WAL, skip-list MemTable, snappy SSTables with sparse index + file-level Bloom so negative lookups skip disk.
- Serializable OCC: snapshot reads, buffered writes, commit-time read-set validation vs a bounded commit log — atomic on no conflict.
- Async leader/follower replication over HTTP NDJSON and gRPC; followers resume from local `LastSeq()` on reconnect, no event replay.

Syncle

- Android spatial-collab client (Kotlin + Compose) renders a 2D office on Canvas with AABB walkability and edge-distance table proximity.
- Realtime over LiveKit: 20 Hz binary position packets, remote-peer lerp, distance-based spatial audio, per-table acoustic isolation.

WeAgree

- Auto-provisions per-user Ed25519 keypair on GitHub OAuth; SimpleWebAuthn passkeys are optional step-up evidence, not the signing key.
- Each signature stores content hash, payload hash, signature, and key fingerprint under Supabase RLS — tamper-evident per-signer audit.
- On finalize, `final_proof_hash` is anchored `bytes32` to Arbitrum Sepolia via Ethers v6; proof API exports an offline-verifier package.

Anchor

- Splunk drift agent fingerprints a healthy window, then ranks diffs against later windows with LLM-narrated SPL suggestions.
- Feedback loop reweights signals and surfaces recurring unresolved drifts as blind spots.

ChefCopilot

- iOS SwiftUI monorepo: on-device CV → SwiftData inventory → FastAPI recipe ranker (feasibility + diversity) → cook-mode step guidance.

CabStream ETL

- Terraform-provisioned GCP stack (GCS + BigQuery + Airflow); whole pipeline tears down and rebuilds from one `apply`.
- dbt three-layer model (staging → dims/facts → marts) with `schema.yml` freshness/uniqueness tests; Airflow DAG logic unit-tested in CI.
- External-table-then-CTAS pattern keeps raw parquet cheap on GCS while marts stay query-optimised in BigQuery.

CERTIFICATES & SKILLS

Coursera: IBM Vector Database, Stanford Machine Learning, UCI Database Design & Operational BI, Rice Parallel-Concurrent-Distributed Programming, UA Reinforcement Learning

Skills: Python, SQL, C#, C++, Java, Git, Go, Kotlin, JavaScript, Linux, Windows, MacOS, .Net, Spring Boot, React, Node.js, Express.js, Flask, PostgreSQL, MySQL, MongoDB, Redis, AWS, GCP, Azure, Docker, Kubernetes, Terraform, GitHub Actions, Jira, Confluence, PyTorch, TensorFlow, Scikit-learn, OpenCV, GenAI tools